

1st Sessional
Deep Learning (BCSML0555A/BCSML0552)
Session: 2026-27 (V Sem.)

Total Marks:30
Section A Quize:10X1=10
Section B Marks :4X5=20
Date:16/07/2026

General Instructions

1. Attempt all four questions. All questions carry equal marks.
2. The question paper consists of two sections:
 - Section A: Objective-type questions.
 - Section B: Programming-based questions with problem statements.
3. Students are required to write, execute, and test the complete program for each problem statement.
4. Clearly display the output of each program. Include intermediate results wherever applicable.
5. Assume appropriate input values, datasets, and parameter values wherever required unless otherwise specified.
6. Students may use any IDE or development platform such as Visual Studio Code (VS Code), Cursor, Jupyter Notebook, PyCharm, Spyder, or any other equivalent platform they are comfortable with.
7. Use appropriate variable names, comments, and proper indentation to improve code readability.
8. Ensure that the program is error-free and executable.
9. Save all source code files properly before submission.

Submission Instructions

After completing all programming questions in **Section B**:

1. Open **Microsoft Word (MS Word)**.
2. Prepare a document containing the following details on the first page:
 - Name
 - Roll Number
 - Branch
 - Section
 - Workshop/Lab Number
3. Copy and paste the **source code** of each program into the document.
4. Insert the corresponding **output screenshots** below each program.
5. Ensure that all screenshots are **clear and readable**.
6. Save the document in **PDF** format.
7. Name the PDF file using the following format:

NAME_ROLLNUMBER_BRANCH_SECTION.pdf

Example:

RahulSharma_2412345678_CSEAIML_A.pdf

8. Upload the PDF file to **Microsoft Forms** before the submission deadline.

Section B**QUESTION 1: Smart Scholarship Eligibility Prediction Using an Artificial Neuron:****[CO1: K4]****Problem Statement**

A university wants to automate the scholarship selection process using an **Artificial Neuron**. The neuron should analyze a student's academic and extracurricular performance and decide whether the student is eligible for a scholarship.

Given Parameters

Parameter	Value
CGPA	0.92
Attendance	0.85
Family Income Score	0.30
Research Activity Score	0.75
Discipline Score	0.95

Initialize the Weights

Input	Weight
CGPA	0.40
Attendance	0.15
Family Income Score	-0.20
Research Activity Score	0.25
Discipline Score	0.30

Bias = -0.25**Activation Function****Use the Step Activation Function****QUESTION 2: Perceptron Learning for Smart Drone Landing Decision: [CO1: K4]****Problem Statement**

An autonomous drone must decide whether it is safe to land based on multiple environmental conditions. Instead of using predefined weights, the drone should learn the decision boundary using the **Perceptron Learning Algorithm**.

Training Dataset

Wind	Rain	Landing Pad	Battery	Obstacle	Target
0	0	1	1	0	1
1	0	1	1	0	0
0	1	1	1	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0

Initialize

- Initial Weights = 0
- Bias = 0
- Learning Rate = 0.2
- Epochs = 10

Tasks

1. Initialize all weights and bias.
2. Implement the Perceptron Learning Rule.
3. Train the perceptron for the given number of epochs.
4. Display the updated weights after training.
5. Test the perceptron using:

Wind	Rain	Landing Pad	Battery	Obstacle
0	0	1	1	0

and print the landing decision.

QUESTION 3: Hebbian Learning for Smart Greenhouse Irrigation Control: [CO1: K4]

Problem Statement

A smart greenhouse monitors different environmental conditions to determine whether the irrigation system should be **ON** or **OFF**. Instead of assigning fixed weights, the system should **learn the relationship between the inputs and the desired output using Hebbian Learning**.

Training Dataset

Soil Moisture	Temperature	Humidity	Sunlight	Plant Condition	Target
1	1	1	1	1	1
-1	1	-1	1	-1	-1
1	-1	1	-1	1	1
-1	-1	-1	-1	-1	-1

Initialize

- Initial Weights = 0
- Learning Rate = 1

Hebbian Learning Rule

$$w_{new} = w_{old} + (Learning\ Rate \times Input \times Target)$$

Tasks

1. Initialize all weights to zero.
2. Read each training pattern one by one.
3. Update the weights using the Hebbian Learning Rule.
4. Display the final weight vector after training.
5. Predict the irrigation decision for the following input:

Soil Moisture	Temperature	Humidity	Sunlight	Plant Condition
1	1	-1	1	1

QUESTION 4: CNN-Based Fashion Product Classification: [CO2: K6]

Problem Statement

Develop a Convolutional Neural Network (CNN) using TensorFlow/Keras to classify fashion products into 10 different categories such as T-shirt, Trouser, Dress, Coat, Sandal, Sneaker, Bag, and Ankle Boot. Display sample images from the dataset before training, preprocess the images by normalizing and reshaping them, train the CNN model, evaluate its classification accuracy, and display the predicted and actual class labels for the test images.

Given Parameters

Parameter	Value
Dataset	Fashion-MNIST
Image Size	28 × 28 pixels
Image Type	Grayscale
Number of Classes	10
Convolution Filters	32
Kernel Size	3 × 3
Pooling Size	2 × 2
Hidden Layer Neurons	128

Activation Function	ReLU
Output Activation	Softmax
Optimizer	Adam
Loss Function	Sparse Categorical Crossentropy
Epochs	5
Validation Split	20%

Tasks

1. Load the Fashion-MNIST dataset.
2. Display at least 10 sample images with their class names.
3. Normalize and reshape the dataset.
4. Design the CNN architecture.
5. Train the model.
6. Evaluate the testing accuracy.
7. Display predictions for the first 10 testing images.

Wishing you all the best